# Phase 5 | Technology Study Roadmap

*A structured learning plan for understanding the stack and preparing to discuss implementation details.*

React/Vite Edition | Repository-grounded study material | Snapshot: 2026-10-24

## 1. Study sequence

| Order | Topic | Target understanding |
|---|---|---|
| 1 | HTTP and REST | Methods, status codes, validation, CORS, JSON contracts and failure responses. |
| 2 | React and Vite | Components, state, context, effects, lazy loading, routing, proxying and build output. |
| 3 | Node.js and Express | Middleware, routers, async handlers, environment configuration and static serving. |
| 4 | MongoDB Atlas and Mongoose | Documents, schema, indexes, uniqueness, connection strings, access control and backups. |
| 5 | JWT and OAuth | Signing vs decoding, expiry, OAuth redirect flow, callback state and secure token storage. |
| 6 | Third-party APIs | Timeouts, quota, cache freshness, retries, provider normalization and attribution. |
| 7 | Flask and Gemini | Flask routes, request payloads, prompt roles, response parsing, latency and safety. |
| 8 | Leaflet and Phaser | Tile/marker rendering vs POI lookup; scenes, game loop, input and collision handling. |
| 9 | Deployment | Vercel rewrites, Render environment, service URLs, SPA fallback and production logs. |

## 2. Five-session study plan

- Session 1 - Sketch the architecture and trace startup locally. Explain how browser/API requests reach Express.
- Session 2 - Trace registration, login, JWT verification, Google OAuth and profile lookup.
- Session 3 - Trace a SACHET alert from feed to readable detail and compare CAP XML with JSON.
- Session 4 - Trace nearby search, Leaflet rendering, the Phaser drill and separate quiz.
- Session 5 - Trace the chatbot request through Flask to Gemini; rehearse security improvements and a mock interview.

## 3. Concepts to define precisely

- Authentication answers who a user is; authorization determines whether that user may perform an action.
- A JWT is signed, not encrypted by default; do not treat its payload as secret.
- CORS is a browser cross-origin policy, not authentication.
- Leaflet draws the map; a POI API supplies place records.
- A cache can improve latency and reduce provider calls, but creates freshness and invalidation trade-offs.
- A system prompt guides model behavior; it is not a substitute for output validation or safety controls.

## 4. Hands-on exercises

- Add an Express health endpoint that reports app and database readiness without leaking connection details.
- Write request/response examples for /api/auth/login, /api/alerts and /api/nearby.
- Draw the OAuth redirect flow and mark where the token appears in the current implementation.
- Mock Geoapify responses and inspect category normalization and distance sorting.
- Add a timeout and controlled failure contract to the Flask-to-Gemini request.
- Identify which UI states distinguish “no active alerts” from “alert provider unavailable.”

Use the repository as the answer source. When describing an improvement, label it as a proposal rather than claiming it already exists.